Case Study: The Change Healthcare Ransomware Incident

Definition: Ransomware is a type of malicious software that creates a situation where access to data is denied until a ransom is paid. It holds data hostage.

How it Happens:

1. Finding Weakness: In the large-scale Change Healthcare attack in 2024, hackers discovered a remote login account that didn’t have two-factor authentication. This was the entry point for the attack.

2. Stealing Data: Once the hackers gained access, they spent several days within the network copying 6 terabytes of data which is comparable to stealing a large number of patients' records.

3. Ransomware Attack: After the data theft, the hackers initiated the ransomware. Immediately, the company's systems were locked, resulting in pharmacies being unable to process prescriptions and doctors not receiving payments.

4. The Ransom Demand: The hackers requested millions of dollars. It was reported that a payment of $22 million was made, yet the criminals disappeared. The stolen data remained unknown, with no information about its destination.

Who is Targeted?

Ransomware attackers choose targets that present easy entry points, where vulnerabilities are easy to exploit. Organisations with poorly configured networks, unpatched software, or open ports are more likely to be targeted by attackers looking for a quick way in.

Consequence: By 2026, it was confirmed that one-third of Americans had their private medical information exposed. In addition to financial losses estimated at $2 billion, the human cost was significant as patients had to pay full prices for essential medications for weeks, and many small medical practices faced bankruptcy while waiting for systems to be restored.

Sources: https://www.aha.org/news/headline/2025-01-27-reports-change-healthcare-cyberattack-exposed-data-190-million-people

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